

Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids: A Latency–Robustness Trade-off Analysis

Jorge Luis Mayorga Taborda, Yessica Africano Rodriguez, Fernando Jimenez

Department of Electrical and Electronics Engineering

Universidad de los Andes

Bogotá, Colombia

jl.mayorga236@uniandes.edu.co, ya.africano1030@uniandes.edu.co,

fjimenez@uniandes.edu.co

Abstract—Frequency estimation in inverter-based resource (IBR) grids must contend with fast ramps, phase jumps, harmonics, ringdown, and out-of-band components that can overlap within a single local event. This paper presents a comparative benchmark of 16 dynamic frequency estimators across 34 test scenarios grouped into 10 disturbance families in a local relay measurement setting. The test set combines compliance-style dynamic tests, including magnitude steps, ramps, modulation, and phase jumps, with IBR-specific stress conditions such as harmonic injection, power-imbalance ringdown, out-of-band interference, and composite multi-event sequences. Each estimator is tuned via scenario-specific parameter tuning and evaluated using RMSE, peak error, trip-risk duration, settling time, and CPU cost. Rankings vary by disturbance family and performance metric: SOGI-PLL achieves the lowest error on ramps, EKF on magnitude steps and out-of-band interference, and RA-EKF on modulation, while the composite IBR event splits RMSE leadership (PLL) from trip-risk leadership (IpDFT). The hardest cases are the multi-event IBR sequence, 60° phase jumps, and noisy ringdown variants. The benchmark shows that compliance-style dynamic tests provide necessary but insufficient validation for protection-oriented frequency measurement in IBR-dominated grids.

Index Terms—Frequency estimation, inverter-based resources, low-inertia systems, relay protection, signal processing, trip-risk assessment.

I. INTRODUCTION

Frequency estimation affects relaying, under-frequency load shedding, disturbance monitoring, and converter supervision in low-inertia grids. In inverter-based-resource (IBR) systems, the estimation problem is not limited to slow electromechanical drift. Converter-dominated dynamics compress the disturbance timescale and can combine fast RoCoF segments, abrupt phase jumps, harmonics, out-of-band components, and short ringdown intervals within the same protection decision window [1]–[3]. Recent IBR studies also show that converter interactions can reshape the observable frequency response itself [4], [5].

These effects stress estimators through different failure modes. Phase discontinuities can be interpreted by loop-based methods as large frequency excursions or cycle slips. Sustained ramps stress the tracking bandwidth of multi-cycle

spectral methods. Harmonic and interharmonic content increases spectral leakage and can bias estimators whose signal model assumes a single dominant sinusoid. Ringdown adds a transient-recovery problem in which average error, peak error, and trip-risk duration need not favor the same method.

The estimator families used in practice reflect this latency–robustness tension. Windowed spectral methods such as IpDFT and Taylor–Fourier variants reject steady harmonic content but pay observation delay [6], [7]. PLL and SOGI-based trackers are light enough for embedded deployment but can lose phase alignment under abrupt discontinuities [8], [9]. Kalman-filter variants can track fast transients when the state model remains valid [10], [11]. Reduced-order and data-driven methods add further cost–accuracy trade-offs, but their behavior under relay-oriented composite stress remains less standardized [12], [13].

This paper frames the contribution as a benchmark study rather than as a single-estimator paper. The benchmark extends standard dynamic-test families with IBR-specific harmonic, out-of-band, ringdown, and multi-event stress in a local, non-time-synchronized setting. The study compares 16 estimators over 34 scenarios using RMSE, peak error, trip-risk duration, settling behavior, and relative CPU cost. The contribution is therefore the disturbance-family comparison protocol and the resulting map of trade-offs; no new estimator family is proposed.

II. RELATED WORK AND ALGORITHMIC LANDSCAPE

This paper addresses local frequency measurement for relay and monitoring functions, not PMU time-stamp quality. IEC/IEEE 60255-118-1 and the earlier IEEE C37.118.1 define dynamic tests around ramps, magnitude steps, modulation, and selected phase events [14], [15]. These tests are necessary reference cases, but they do not by themselves describe how estimators behave when phase discontinuities, harmonics, ramps, and ringdown occur in the same local record.

Windowed spectral methods, PLL variants, Kalman filters, and reduced-order or data-driven estimators each occupy a different point on the latency–robustness plane. IpDFT and

TFT are not interchangeable. IpDFT uses a stationary spectral model with a Hanning window and parabolic interpolation around the dominant bin [6]. TFT uses a second-order Taylor-series expansion of the fundamental component only, also under Hanning weighting, which gives explicit phasor-derivative tracking at the cost of higher noise sensitivity [7]. PLL variants are computationally light but can misread abrupt phase steps [8], [9]. EKF- and UKF-based approaches can improve transient tracking when the state model matches the event, and RoCoF augmentation in EKF-type estimators has been studied previously [10], [11]. Reduced-order and data-driven methods add further cost trade-offs [12], [13].

Prior low-inertia and IBR studies show why this gap matters [1]–[5]. The remaining need is a comparison that reports standard dynamic tests, composite IBR stress, trip-risk, and compute cost in the same frame. The benchmark used here keeps the standard disturbance families, adds IBR-specific harmonic, ringdown, out-of-band, and multi-event stress, and reports family-level summaries rather than a single global winner. This framing targets protection-oriented estimator selection, not synchrophasor compliance certification.

III. METHODOLOGY AND TEST FRAMEWORK

The benchmark compares 16 frequency estimators on a common dynamic test set in a local, non-time-synchronized measurement setting, using waveform generation at 1 MHz and estimator outputs at 10 kHz. The scenario set extends the same protection-oriented stress modes used in standard dynamic testing—magnitude steps, ramps, modulation, and phase jumps—with IBR-specific harmonic, out-of-band, ringdown, and multi-event stress. This expansion does not change the measurement setting: all methods are evaluated as local relay-class frequency trackers from the same decimated 10 kHz data stream.

The estimator pool spans six practical groups: zero-crossing and energy baselines (ZCD, TKEO), windowed spectral methods (IpDFT, TFT), PLL/SOGI-based trackers (PLL, SOGI-PLL, SOGI-FLL, Type-3 SOGI-PLL), Kalman-filter methods (LKF2, EKF, UKF, RA-EKF), parametric/subspace methods (Prony, ESPRIT), and reduced-order or adaptive/data-driven designs (RLS, Koopman/RK-DPMU). This mix is intended to expose trade-offs relevant to protection and monitoring: accuracy, transient risk, settling behavior, and computational burden. Within the Kalman-filter group, EKF and UKF use a single-tone state over phase, frequency, and amplitude, whereas RA-EKF adds an explicit RoCoF state, innovation-scaled process noise, and a soft phase reset under large innovations. RoCoF augmentation in EKF-type filters has prior literature antecedents [10], [11]; the benchmark treats RA-EKF as one Kalman-family variant, not as a new estimator family.

The study contains 34 benchmark scenarios. Rather than read the benchmark as a flat scenario list, we group the scenarios by disturbance family. Table I summarizes the resulting ten-family structure, and Fig. 1 shows representative ground-truth signals for six of those families.

The family structure reflects distinct physical stress modes. Ramps probe sustained RoCoF tracking. Magnitude steps

TABLE I
BENCHMARK SCOPE. COUNTS REFER TO BENCHMARK SCENARIOS. OOB DENOTES OUT-OF-BAND INTERFERENCE. THE PHASE-JUMP FAMILY INCLUDES 20° AND 60° RE-LOCK TESTS.

Family	Count	Stress tested
IEEE ramps	9	Sustained RoCoF from 0.25 to 20 Hz/s
Mag. steps	7	1% to 50% magnitude changes
Modulation	3	Nominal, AM, and FM tracking
Phase jumps	3	20° and 60° phase-jump re-lock tests
OOB interference	1	Spectral rejection under off-nominal content
Single-tone	1	Deterministic steady-state control case
IBR harmonics	3	Small, medium, and large harmonic stress
IBR ringdown	5	Power-imbalance ringdown with noise sweep
IBR multi-event	1	Composite phase jump, ramp, harmonics, and ringdown
Freq. step	1	Isolated frequency-step control case

and modulation probe frequency-amplitude decoupling. Phase jumps probe re-lock behavior and transient overshoot. Out-of-band interference and IBR harmonics probe spectral leakage. Ringdown and multi-event cases combine these effects under changing noise and transient recovery, which makes them the strongest stress tests in the benchmark. Two records drive the sharpest separations: a NERC-type +60° phase-jump case at $t = 0.7$ s with 4% 5th, 2% 7th, and 0.5% 32.5-Hz interharmonic content, and an IBR multi-event case with a 0.5-pu sag/swell, a +45° phase jump, a −2 Hz/s RoCoF ramp clamped at 58 Hz, decaying DC, AM/PM oscillations, a full IBR harmonic spectrum, and compound noise. Figure 1 illustrates that difference directly: some benchmark families are isolated compliance-style events, whereas others stack abrupt discontinuities, ramps, and oscillatory recovery inside the same record.

All estimators are evaluated as local relay-class frequency trackers from the same decimated 10 kHz data stream; the study does not assess PMU time-stamping or synchrophasor compliance. Each estimator–scenario pair is tuned via scenario-wise parameter search across 100 trials and then evaluated over 60 Monte Carlo runs. The reported values therefore approximate best-in-range performance within the tested search space, not untuned default settings. The study contains 544 estimator–scenario summaries derived from 32,640 Monte Carlo records. Let $e[k] = \hat{f}[k] - f_{\text{true}}[k]$. We report RMSE = $\sqrt{N^{-1} \sum_k e[k]^2}$, maximum peak error $\max_k |e[k]|$, trip-risk duration $T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$, settling time, and CPU time per sample. The 0.5-Hz threshold is used as a protection-oriented risk proxy rather than as a universal trip criterion. CPU time is interpreted as a relative computational-cost indicator measured in one software environment, not as hardware latency. For any metric m , the family summary is $\bar{m}_F = |F|^{-1} \sum_{s \in F} \bar{m}_s$, where $\bar{m}_s$ is the scenario mean over the 60 realizations. This prevents dense families from dominating the comparison. Unless stated otherwise, frequency errors are reported in mHz in the tables and discussion.

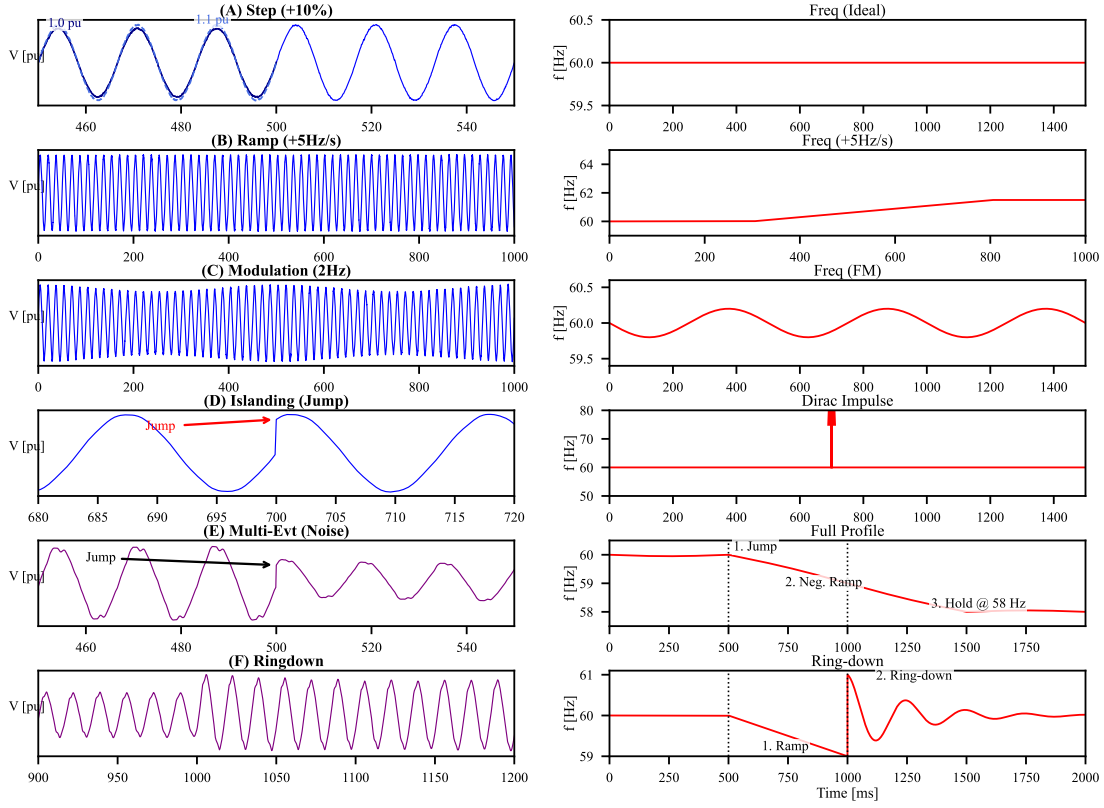


Fig. 1. Representative benchmark scenarios. Each row shows the voltage waveform and true frequency trajectory for six disturbance families: 10% magnitude step, frequency ramp, modulation, 60° phase jump, IBR multi-event sequence, and IBR power-imbalance ringdown. Dashed lines mark disturbance onset. The figure illustrates why the benchmark is read by disturbance family: some cases isolate standard dynamic stress, whereas others combine discontinuities, ramps, harmonics, and oscillatory recovery in one record. Panel (E) shows the dominant frequency profile of the multi-event sequence; the full scenario also includes a 0.5-pu sag/swell, AM/PM modulation, a full IBR harmonic spectrum, and compound noise.

TABLE II

CORE DISTURBANCE-FAMILY COMPARISON FOR THE BENCHMARK STUDY. RMSE VALUES ARE REPORTED IN mHz AND COMPUTED AS FAMILY MEANS OF SCENARIO MEANS. THE UNCERTAINTY TERM IS THE MEAN WITHIN-SCENARIO MONTE CARLO STANDARD DEVIATION FOR THE WINNING ESTIMATOR. Δ IS THE RMSE GAP BETWEEN THE WINNER AND THE RUNNER-UP.

Family	Count	RMSE winner	RMSE [mHz]	Runner-up	Δ [mHz]	Lowest T_{trip} [s]
IEEE ramps	9	SOGI-PLL	19.4 ± 6.4	Koopman	2.7	0 (multiple)
Mag. steps	7	EKF	0.388 ± 0.143	RA-EKF	1.45	0 (multiple)
Modulation	3	RA-EKF	17.8 ± 21.4	Koopman	13.7	0 (multiple)
Phase jumps	3	SOGI-FLL	174 ± 67.5	PLL	4.03	ZCD: 0.0194
OOB interf.	1	EKF	13.6 ± 11.1	RA-EKF	7.35	EKF, RA-EKF: 0
IBR harmonics	3	EKF	70.5 ± 27.3	TFT	4.13	TFT, Prony: 0.00947
IBR ringdown	5	UKF	313 ± 23.2	EKF	6.67	ESPRIT: 0.0989
IBR multi-event	1	PLL	173.5 ± 66.7	UKF	2.74	IpDFT: 0.0439

IV. SIMULATION RESULTS AND BENCHMARKING

A. Family-Level Accuracy and Trip-Risk

Table II summarizes family-level RMSE and trip-risk results. The ranking changes across disturbance families. SOGI-PLL gives the lowest mean RMSE on IEEE ramp cases (19.4 mHz). EKF leads magnitude steps (0.388 mHz) and out-of-band interference (13.6 mHz), and RA-EKF leads the modulation family (17.8 mHz). Zero-trip-risk ties are common in ramps, magnitude steps, and modulation, so those families are separated mainly by accuracy rather than by protection-oriented risk.

Across the 34 benchmark scenarios, nine different estimators win at least one RMSE scenario, and no estimator wins more than 10 of them. The magnitude-step family is the clearest exception: EKF wins all seven magnitude-step scenarios and the family mean as well. This is also the family with the clearest aggregate margin: EKF leads RA-EKF by 1.45 mHz, while the winner's mean within-scenario spread is 0.143 mHz.

Most other families are tighter. In ramps, modulation, phase jumps, harmonics, ringdown, and out-of-band interference, the family-level margin is smaller than or comparable to the same

spread. For those families, the winner should be interpreted as a near-front method rather than as a decisive leader.

Trip-risk is less discriminating in the clean standard cases. In 23 of the 34 scenarios, the minimum trip-risk is zero, often shared by 10 to 13 estimators. Phase jumps are the main exception. SOGI-FLL gives the lowest mean RMSE across the phase-jump family (174 mHz), but ZCD gives the lowest mean trip-risk (0.0194 s). Even inside the standard test set, RMSE and trip-risk are not interchangeable objectives.

B. Composite IBR Stress Cases

The largest ranking changes appear under composite stress. The hardest benchmark scenarios, measured by median RMSE across estimators, are the IBR multi-event sequence (826 mHz), the two 60° phase-jump cases (675 mHz and 637 mHz), and the noisy IBR ringdown variants (474–528 mHz). Median RMSE in these cases exceeds the magnitude-step family by roughly three orders of magnitude.

The IBR multi-event sequence is the clearest example. PLL gives the lowest RMSE at 173.5 mHz. UKF trails by only 2.74 mHz on the family mean. EKF remains close at 176.4 mHz while reducing trip-risk to 0.0874 s. IpDFT gives the lowest trip-risk (0.0439 s) but at the cost of a much larger peak error (2550 mHz). The ranking therefore depends on the deployment objective: minimizing average error, minimizing time above the trip threshold, or bounding transient severity.

The same metric split appears in the other IBR-specific families. EKF gives the lowest mean RMSE across the harmonic family (70.5 mHz), but TFT and Prony give the lowest mean trip-risk there (0.00947 s). In the ringdown family, UKF gives the lowest mean RMSE (313 mHz), whereas ESPRIT gives the lowest mean trip-risk (0.0989 s) at a much higher computational cost.

Among the 12 scenarios with a unique trip-risk winner, nine differ from the RMSE leader. Those disagreements concentrate in the 60° phase-jump cases and in the harmonic, ringdown, and multi-event IBR cases. This metric split is why a single scalar ranking is insufficient for protection-oriented selection.

C. Global Aggregation and Metric Dependence

Family-level reading remains the primary analysis, but global summaries still help as secondary evidence. Their interpretation depends on the aggregation rule. Mean-rank summaries across the 34 benchmark scenarios favor TFT on RMSE (4.38) and peak error (4.32), with RA-EKF close behind on both (4.62 and 4.59). Simple averaging over the 34 scenarios instead favors RA-EKF and EKF, with mean RMSE values of 116 mHz and 123 mHz, followed by UKF at 131 mHz. The difference is informative: mean rank rewards repeated near-front finishes, whereas raw averaging is driven more strongly by the hardest families.

Consistency statistics make that distinction concrete. EKF is the most frequent accuracy leader, with 10 outright RMSE wins, 19 RMSE top-three finishes, and 20 peak-error top-three finishes across the 34 scenarios. TFT wins only two RMSE scenarios, yet it reaches the RMSE top three in 16 scenarios

TABLE III
COMPACT GLOBAL SCREENING INDICATORS ACROSS THE 34 BENCHMARK SCENARIOS. THESE STATISTICS SUMMARIZE DIFFERENT AGGREGATION RULES AND ARE INTENDED AS SCREENING AIDS RATHER THAN AS A UNIVERSAL RANKING.

Global view	Best estimator	Value
Mean RMSE	RA-EKF	116 mHz
Mean-rank RMSE	TFT	4.38
RMSE wins	EKF	10 / 34
RMSE top-three	EKF	19 / 34
Peak-error top-three	EKF	20 / 34
Trip-risk top-three	IpDFT	25 / 34

and the peak-error top three in 17. Trip-risk favors a different group again: IpDFT and ESPRIT enter the trip-risk top three in 25 and 24 scenarios, respectively. Global summaries are therefore screening tools rather than replacements for family-level evidence.

Table III makes the aggregation split explicit. The rank-based view favors repeated near-front finishes, the mean-RMSE view is more sensitive to the hardest families, and the hit-count view rewards broad coverage. No estimator leads all three views at once. Across the 544 estimator–scenario summaries, RMSE correlates strongly with trip-risk ($\rho = 0.890$) and settling time ($\rho = 0.867$), but only weakly and negatively with CPU cost ($\rho = -0.202$).

D. Computational Cost

CPU cost changes the practical ranking again. Figure 2 gathers representative time-domain cases together with the global summaries. EKF and RA-EKF occupy the low-error middle-cost region with mean CPU costs of $0.813 \mu\text{s}$ and $1.61 \mu\text{s}$ per sample. SOGI-PLL and SOGI-FLL remain much cheaper ($0.0867 \mu\text{s}$ and $0.0348 \mu\text{s}$) and therefore stay relevant for embedded protection even when they lose some accuracy. Window-based methods such as TFT and IpDFT improve trip-risk in some composite cases at about $3\text{--}4 \mu\text{s}$ per sample. Koopman and ESPRIT occupy a much higher-cost region, with mean costs of $390 \mu\text{s}$ and $2838 \mu\text{s}$, without a matching global RMSE gain. On the RMSE–CPU plane, EKF, RA-EKF, SOGI-FLL, SOGI-PLL, and ZCD are Pareto-optimal; in practice, the protection-relevant subset is {SOGI-FLL, SOGI-PLL, EKF, RA-EKF}, since ZCD reaches the front only by near-zero cost and not by usable composite-stress accuracy.

For deployment screening within this benchmark, estimator choice remains tied to disturbance family, operating objective, and compute budget. SOGI-PLL is strongest on ramps, EKF on magnitude steps and out-of-band interference, RA-EKF on modulation, and PLL on the composite IBR event. Figure 2(g) keeps SOGI-FLL, SOGI-PLL, EKF, and RA-EKF in the main low-cost cluster below $2 \mu\text{s}$ per sample, whereas TFT and IpDFT move rightward in cost but improve trip-risk behavior in the composite cases. Those are conditional results, not a universal hierarchy.

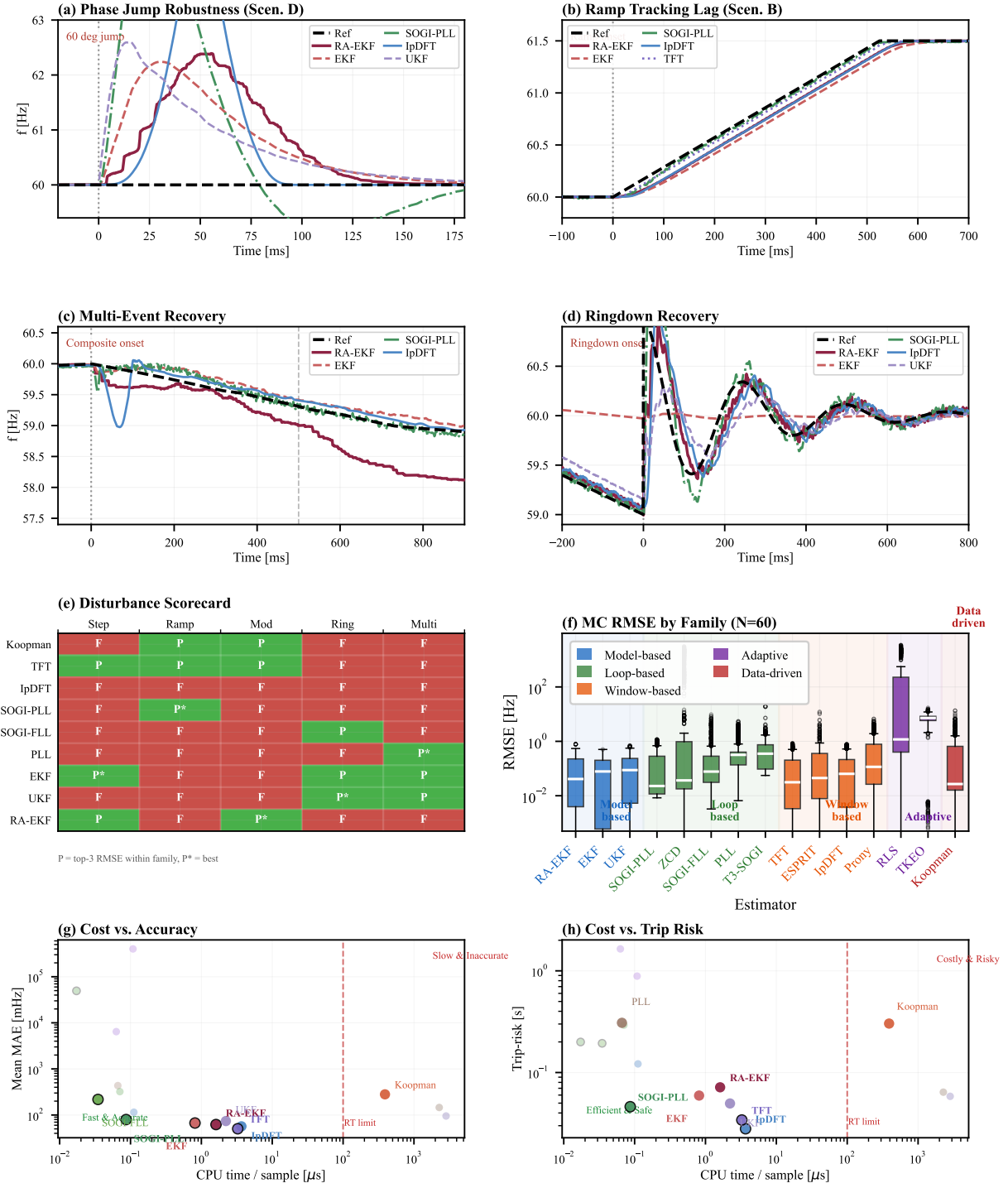


Fig. 2. Representative benchmark views. Panels (a)–(d) show single-scenario responses summarized over the same 60 Monte Carlo runs used throughout the study. Panel (e) gives a disturbance scorecard across benchmark families. Panel (f) shows the scenario-level RMSE distribution grouped by estimator family. Panels (g) and (h) show global cost-versus-accuracy and cost-versus-trip-risk views; the dashed vertical line in both panels marks the 100- μ s/sample real-time budget implied by the 10-kHz acquisition rate. The dashboard shows that estimator selection depends on disturbance family, risk metric, and compute budget.

V. CONCLUSION

This paper compares 16 frequency estimators over 34 scenarios in a local relay-class measurement setting that combines standard dynamic tests with IBR-specific composite stress. The results show disturbance dependence: estimator rankings

move with both the disturbance family and the metric used to read them. Standard compliance-style tests remain necessary for protection-oriented evaluation in IBR-dominated grids.

We highlight three findings. First, the latency–robustness

trade-off is structural: nine estimators win at least one RMSE scenario, and no estimator wins more than 10 of the 34. Windowed spectral methods reduce steady harmonic error but pay observation delay, while low-cost loop-based methods remain attractive under tight computational budgets. Second, Kalman-family methods, especially EKF and RA-EKF in this dataset, remain competitive across the widest range of families. RA-EKF is evaluated here as one Kalman-family variant rather than as a new estimator family. Third, the hardest cases are the IBR multi-event sequence, the 60° phase-jump cases, and the noisy ringdown variants; these are also the cases in which RMSE, peak error, and trip-risk select different estimators.

The study is limited to the scenario set and relative computational-cost measurements reported here. CPU time is used as a software-level cost indicator, not as hardware latency, and the study remains simulation-only. Hardware-in-the-loop validation, wider coverage of converter-control interactions, and sensitivity analysis of the tuning protocol are the natural next steps.

From a deployment standpoint, the benchmark identifies a small practical core. SOGI-PLL is the strongest low-cost choice for ramp-dominated operation. EKF is the most consistent model-based baseline for magnitude steps, out-of-band interference, and balanced composite-event behavior. RA-EKF earns its place not as a universal winner but as a higher-cost Pareto point with the best mean RMSE and the strongest modulation-family results. IpDFT and TFT remain relevant when limiting trip-risk matters more than minimizing average error, provided their observation delay is acceptable.

In practice, the benchmark does not support a single default estimator for all low-inertia IBR conditions. It supports a screening workflow: narrow candidates by disturbance family and compute budget, then resolve the final choice with the metric aligned to the protection objective. In that workflow, ramps mostly test tracking bandwidth, jumps test recovery after discontinuities, and harmonic-rich cases test robustness to leakage and interharmonics.

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